

Layer-2 Overlay Network over WireGuard

0.5cm] VPS Proxmox Virtualization Environment

1.5cm]

Technical Architecture & Implementation Document

0.5cm] *Author : Jacek Kaplon*

Year: 2026

3cm]

Contents

1 Executive Summary

This document describes the design and implementation of a secure, encrypted Layer-2 (L2) overlay network between a public VPS and a Proxmox hypervisor hosted on a private FTTH connection. The solution uses WireGuard as the Layer-3 transport, GRETAP for Ethernet encapsulation, and Linux bridges to merge both sites into a single broadcast domain.

The overlay enables virtual machines on Proxmox to behave as if they were physically located inside the VPS network, supporting ARP, DHCP, multicast, and broadcast traffic. The system is automated using systemd and is production-ready.

2 Project Objectives

Primary Goals

- Build a stable, encrypted L3 tunnel between VPS and Proxmox.
- Implement an L2 overlay on top of WireGuard.
- Allow VMs to obtain IP addresses from the VPS network.
- Provide full broadcast domain extension.
- Ensure automatic startup after reboot.
- Maintain clean, maintainable, production-grade configuration.

Secondary Goals

- Provide reusable automation scripts.
- Create a portfolio-ready engineering case study.

3 System Overview

The system consists of:

- VPS (public IP, Debian/Ubuntu)
- Proxmox VE (private FTTH)
- WireGuard (L3 encrypted tunnel)
- GRETAP (L2 encapsulation)

- Linux bridges (L2 broadcast domain)
- NAT for VM Internet access
- systemd automation

4 Architecture

4.1 Physical Layer

[Proxmox Host] ---- FTTH ---- Internet ---- [VPS]

4.2 Layer-3 Transport (WireGuard)

```
10.77.77.2 (Proxmox)
    encrypted UDP/51830
10.77.77.1 (VPS)
```

4.3 Layer-2 Encapsulation (GRETAP)

```
gretap-overlay (Proxmox)
    Ethernet-over-WireGuard
gretap-overlay (VPS)
```

4.4 Overlay Bridge

```
Proxmox: vmbr-overlay
VPS:      br-overlay
```

4.5 Addressing Plan

Component	Address
WireGuard VPS	10.77.77.1/24
WireGuard Proxmox	10.77.77.2/24
Overlay Subnet	10.50.0.0/24
VPS Bridge	10.50.0.1
Proxmox Bridge	10.50.0.2

5 Implementation

5.1 VPS Configuration

WireGuard

```
[Interface]
Address = 10.77.77.1/24
ListenPort = 51830
PrivateKey = <private>

PostUp = sysctl -w net.ipv4.ip_forward=1
PostDown = sysctl -w net.ipv4.ip_forward=0

[Peer]
PublicKey = <Proxmox-public>
AllowedIPs = 10.77.77.2/32
```

GRETAP

```
ip link add gretap-overlay type gretap \
    remote 10.77.77.2 local 10.77.77.1 dev wg-overlay
ip link set gretap-overlay up
```

Bridge

```
ip link add br-overlay type bridge
ip link set br-overlay up
ip link set gretap-overlay master br-overlay
ip addr add 10.50.0.1/24 dev br-overlay
```

5.2 Proxmox Configuration

WireGuard

```
[Interface]
Address = 10.77.77.2/24
PrivateKey = <private>

[Peer]
PublicKey = <VPS-public>
```

```
Endpoint = <VPS-IP>:51830
AllowedIPs = 10.77.77.1/32
PersistentKeepalive = 25
```

GRETAP

```
ip link add gretap-overlay type gretap \
    remote 10.77.77.1 local 10.77.77.2 dev wg-overlay
ip link set gretap-overlay up
```

Bridge

```
ip link add vmbr-overlay type bridge
ip link set vmbr-overlay up
ip link set gretap-overlay master vmbr-overlay
ip addr add 10.50.0.2/24 dev vmbr-overlay
```

6 Automation (systemd)

6.1 Overlay Startup Script

```
/usr/local/bin/overlay-vps.sh
```

6.2 Overlay Service

```
[Unit]
Description=Overlay Network (GRETAP + Bridge)
After=wg-quick@wg-overlay.service
Requires=wg-quick@wg-overlay.service

[Service]
Type=oneshot
ExecStart=/usr/local/bin/overlay-vps.sh
RemainAfterExit=yes

[Install]
WantedBy=multi-user.target
```

7 Validation & Testing

WireGuard Handshake

```
wg show
```

L3 Connectivity

```
ping 10.77.77.1  
ping 10.77.77.2
```

L2 Connectivity

```
ping 10.50.0.1  
ping 10.50.0.2
```

8 Troubleshooting & Debugging Notes

- Incorrect WireGuard private key caused handshake failure.
- GRETAP failed to attach due to missing bridge.
- Netplan incompatible with advanced overlay networking.
- tcpdump used to confirm UDP packet flow.

9 Security Considerations

- WireGuard uses modern cryptography (Curve25519, ChaCha20-Poly1305).
- All L2 traffic is encrypted.
- NAT prevents unsolicited inbound traffic.

10 Performance Considerations

- GRETAP adds overhead; MTU reduced to 1382.
- WireGuard is kernel-level and extremely fast.
- Suitable for VM workloads and overlay services.

11 Future Enhancements

- VXLAN support.
- BGP dynamic routing (FRR).
- DHCP relay for VM auto-provisioning.
- Multi-VPS redundancy.
- Monitoring (Prometheus + Grafana).

12 Lessons Learned

- WireGuard is extremely sensitive to key mismatches.
- Netplan is unsuitable for complex overlay networking.
- systemd is the correct tool for automation.
- tcpdump is essential for debugging tunnels.